

System Run Instructions

This document provides step-by-step instructions to configure, run, and verify the application in a local or containerized environment.

1. System Prerequisites

- Node.js: v20.x, v22.x, or newer installed
- Git: Installed and available in system PATH
- PowerShell or Terminal: Standard command shell
- Redis: Included portably in bin/redis/redis-server.exe (Windows) or external Redis

2. Environment Configuration (.env.local)

Ensure a .env.local file exists in the project root directory with the following variables:

```
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=your-anon-key
SUPABASE_SERVICE_ROLE_KEY=your-service-role-key
REDIS_URL=redis://127.0.0.1:6379
OPENAI_API_KEY=your-api-key
OPENAI_BASE_URL=https://api.openai.com/v1
LLM_MODEL=gpt-4o-mini
GITHUB_TOKEN=ghp_yourPersonalAccessTokenHere
```

3. Execution Method A: Manual 3-Terminal Setup

The platform consists of three coordinating processes. Open three separate terminal windows in the project root directory:

Step 1: Start Redis Server (Terminal 1)

Run the portable Redis binary:

```
.\bin\redis\redis-server.exe
```

Verify Redis is responding by running in another prompt: `.\bin\redis\redis-cli.exe ping` (should return PONG).

Step 2: Start Background Worker (Terminal 2)

Run the queue worker that handles background scanning jobs:

```
npm run worker
```

You should see: Connected to Redis and ready to process jobs.

Step 3: Start Web Server (Terminal 3)

Launch the Next.js development server:

```
npm run dev
```

Open your web browser and navigate to: `http://localhost:3000`

4. Execution Method B: 1-Click Windows Batch Launcher

A single-click launcher is available in the root folder:

- Double-click start-all.bat or run `.\start-all.bat` in PowerShell.
- This script checks Redis connectivity, starts redis-server if not running, launches worker.ts in a dedicated window, and starts npm run dev in another window.

5. Execution Method C: Docker Compose

To run Redis, the security engine toolchain, and the application in isolated containers:

```
docker compose up --build -d
```

The web application will be accessible at <http://localhost:3000>.

To stop containers: `docker compose down`

6. Verification & Troubleshooting

- Redis Connection: Test with `.\bin\redis\redis-cli.exe ping`. Expected reply: PONG.
- Web Server: Open <http://localhost:3000> in your browser to verify the dashboard loads.
- Worker Status: The worker console should display job receipt and completion logs when a scan is initiated.
- Port Conflicts: Ensure port 6379 (Redis) and port 3000 (Web) are not occupied by other services prior to startup.